

ECON 100B: Macroeconomics

Class 16: The IS Curve (Jones Chapter 11)

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March 17, 2026

Agenda

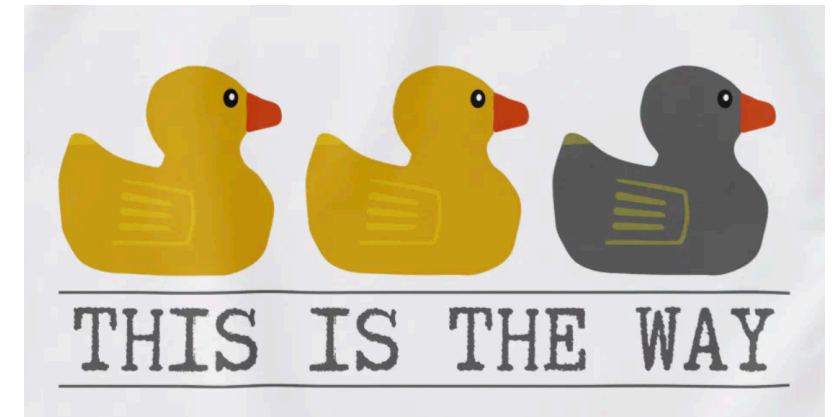
- Announcements
- Current events
- Recap of takeaways from Jones Chapter 10: Recent and not-so-recent Macro-clysms
- Chapter 11: The IS Curve

Announcements

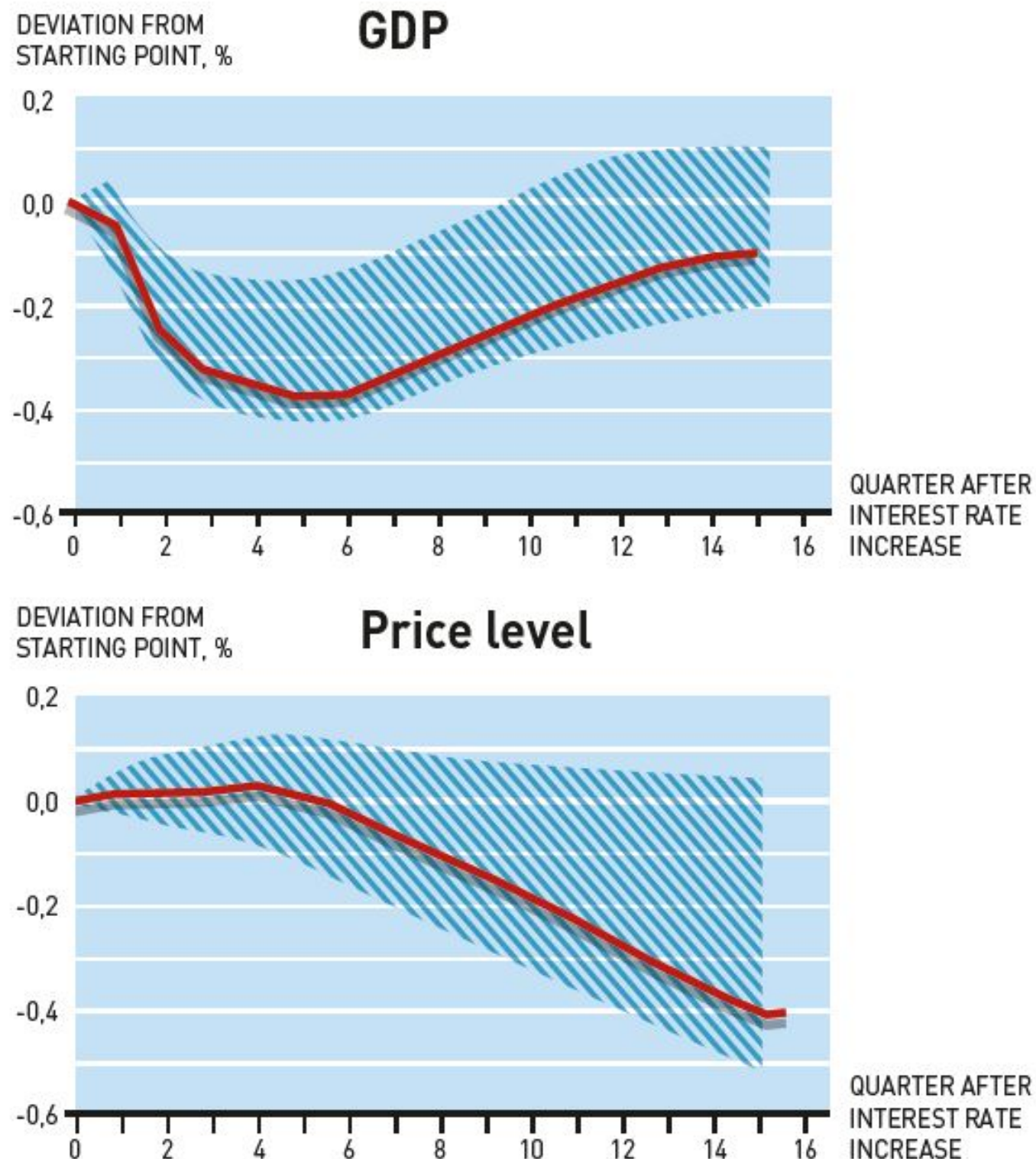
- Take-home Midterm Essay released 3/8, due 3/20
 - Scores will be high
 - You can use many tools including AI, but essays written solely by AI typically lose points and break the rules. Up to 10% of ECON 157 for example. AI is not a student in this classroom and won't know the best answers
- Spring Break 3/23-3/27

2011 Nobelist Chris A. Sims (1942-2026)

- Chris Sims died last week at age 83
- Minnesota 1970-1990
Yale 1990-1999
Princeton 1999-emeritus
- Shared the Nobel in 2011 with Tom Sargent (NYU)
- Sims advanced the field of **empirical macroeconomics**
 - “Vector autoregressions” which you might see in ECON 140/141
 - Causal inference in macroeconomics is pretty difficult, and VARs were an improvement over what came before



From the Nobel page, a monetary policy VAR



- In ECON 100B, theory reveals the short-run IS-MP model
- But what do the data say about the effects of a 1 percentage point increase in the target federal funds interest rate?
- This is related to:

“What is the actual slope of the IS curve?”

AND

“What is the actual slope of the Phillips curve?”

- The VAR method reveals pretty good answers consistent with later methods, and it reveals how the **timing** is interesting

Current events

- Rising oil prices
- [As Justin Wolfers \(UMichigan\) discusses](#), U.S. consumers are hurt while U.S. oil companies and oil workers are helped
- [Wolfers also posts](#) markets expect tighter Fed monetary policy = higher federal funds interest rate
- [On 3/18/26, the Fed announced](#) no change in interest rates. The vote was 11-1 (Miran)



Justin Wolfers ✓ @justinwolfers.bsky.social · 2h

Worth noting just how much the Iran war has changed expectations about the Fed and the future path of monetary policy.

War in Iran has led to expectations of higher interest rates

Expected future interest rates (Fed funds futures)

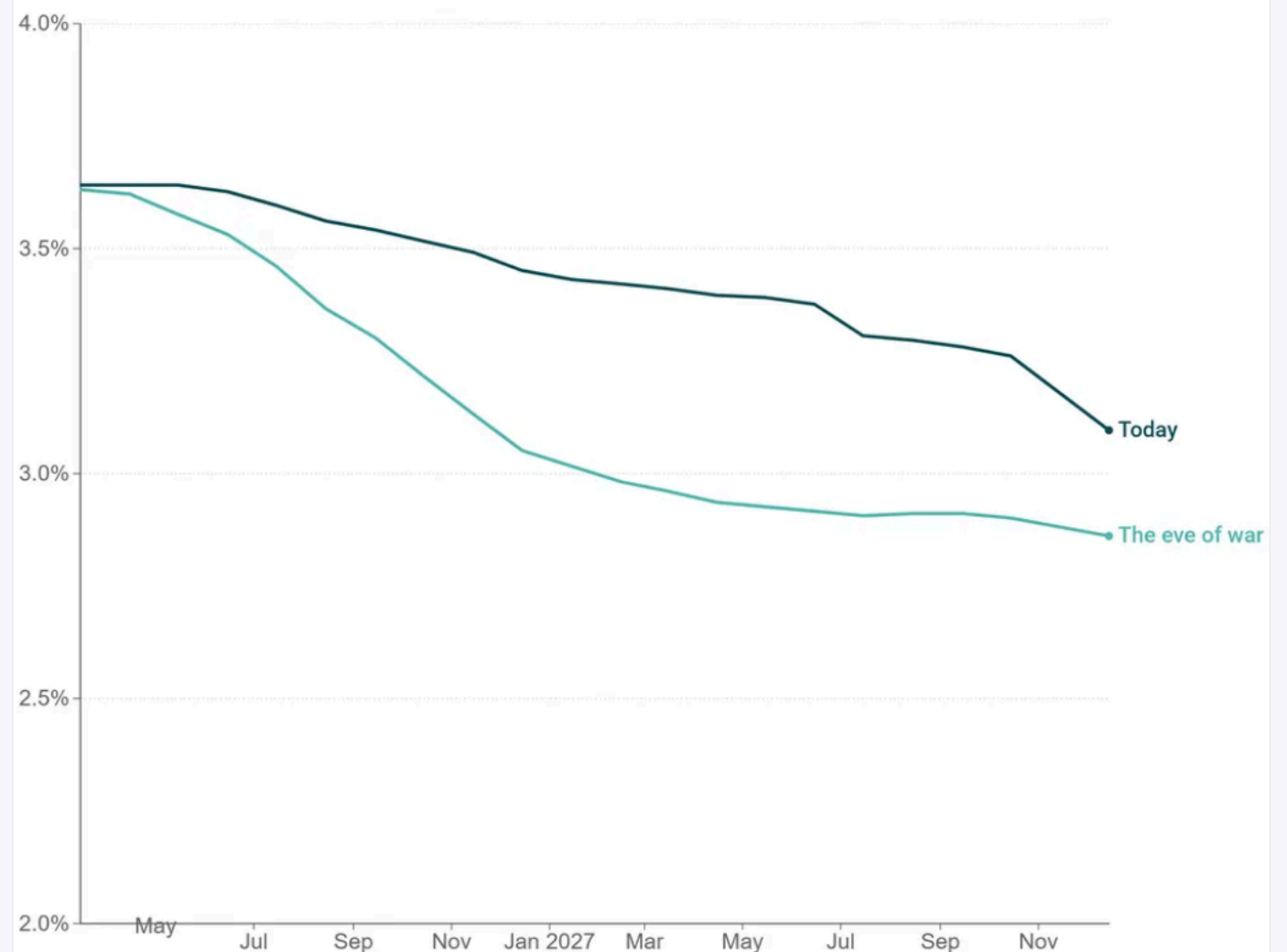
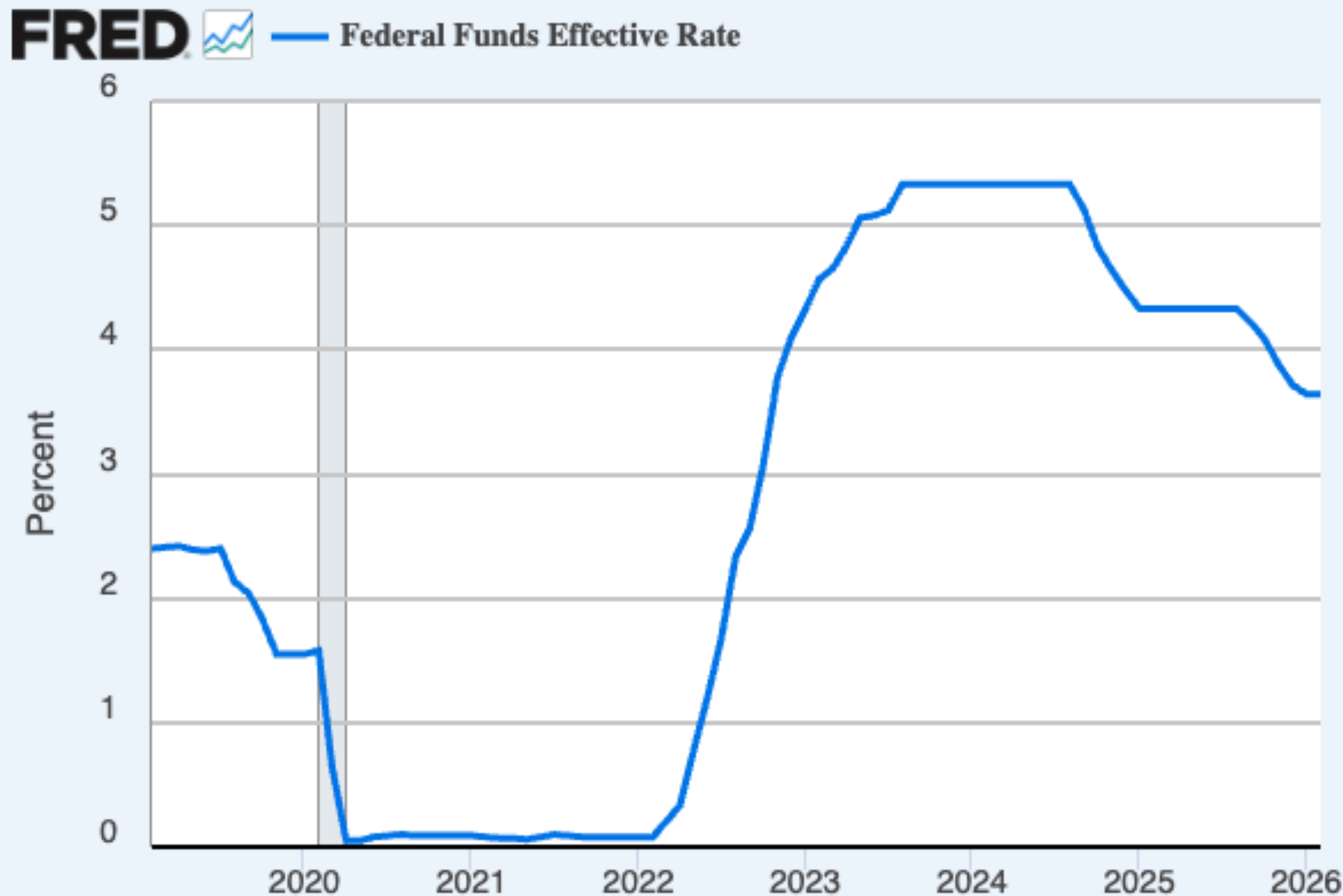


Chart: @JustinWolfers • Source: Yahoo Finance • Series: ZQ • Created: Mar 18, 2026

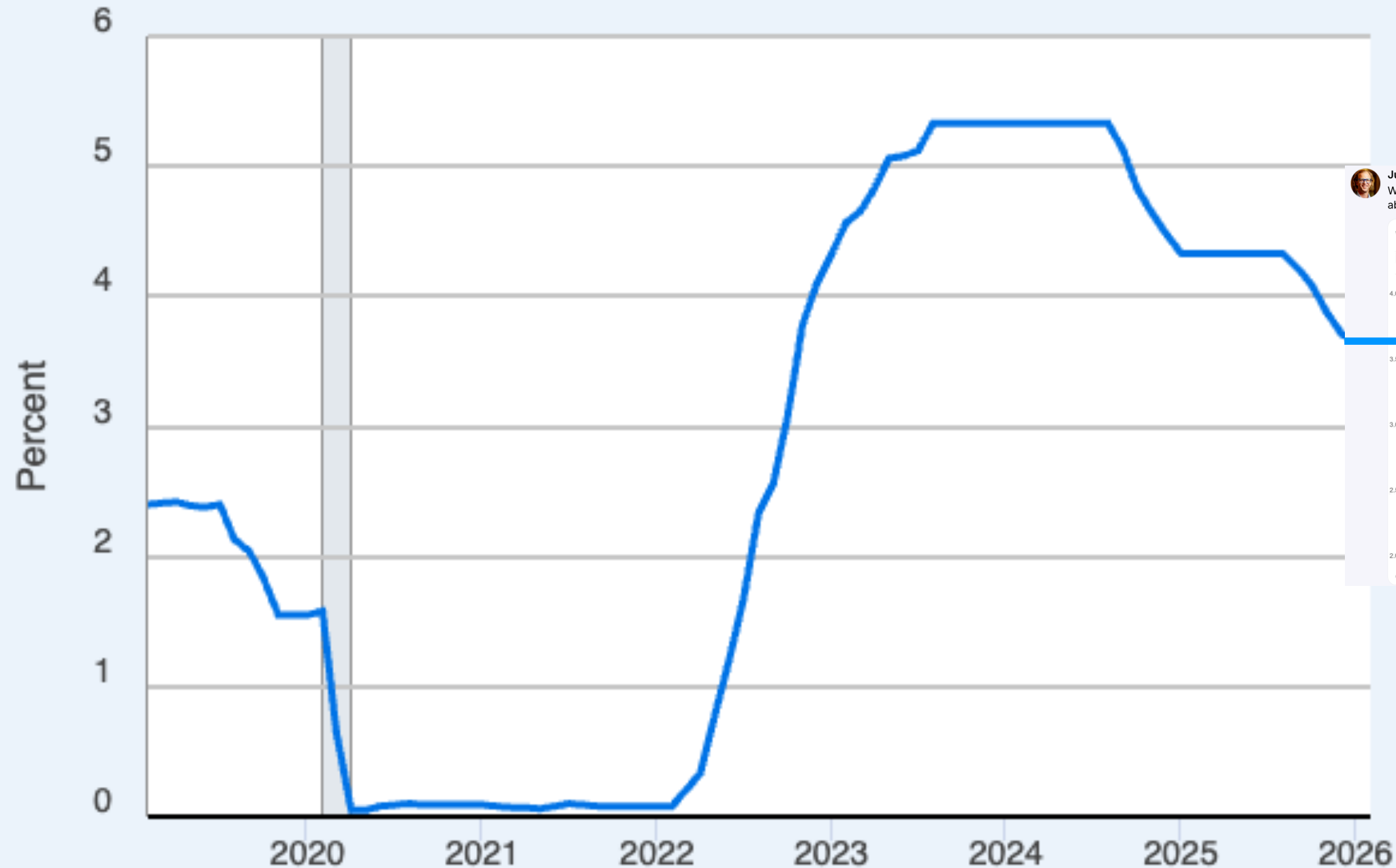
The federal funds rate is the primary tool of the Federal Reserve. It is currently 3.64% with a policy range of 3.50-3.75%



Source: Board of Governors of the Federal Reserve System (US) via FR...

The federal funds rate is the primary monetary policy tool of the Federal Reserve

FRED  — Federal Funds Effective Rate



Source: Board of Governors of the Federal Reserve System (US) via FR...

Market participants reveal their expectations of future Fed policy in several ways. [One is trading in Kalshi markets](#)



Economics · Fed

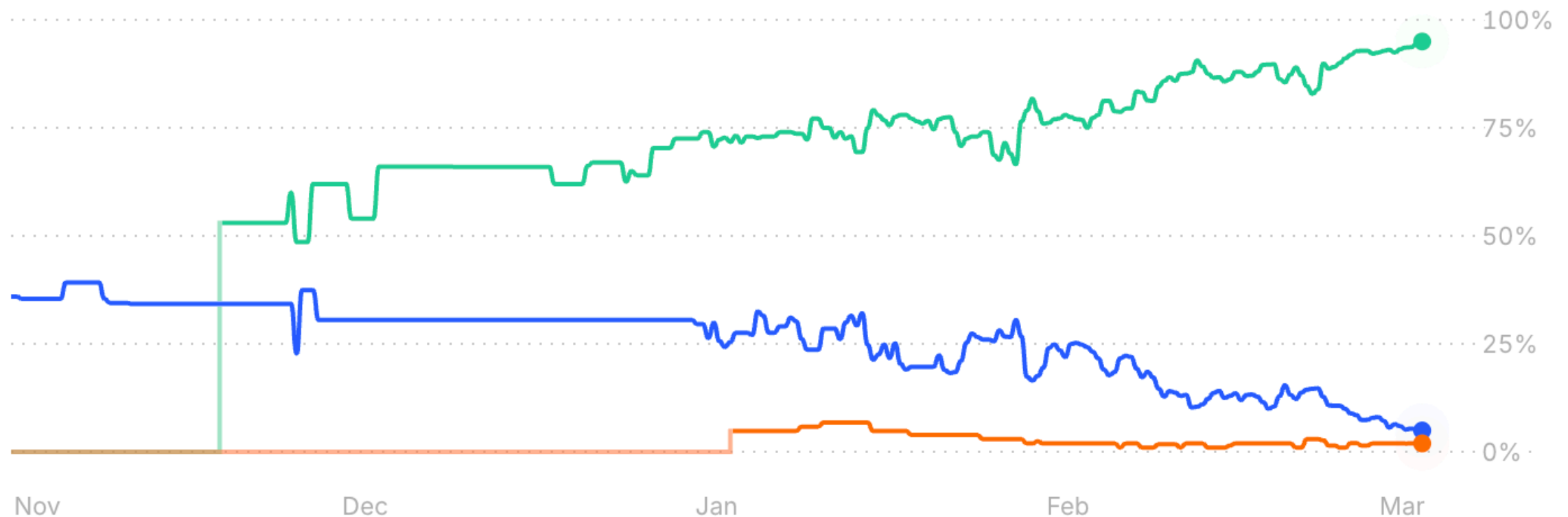
Fed decision in April?



Begins in 42 days · Apr 29, 1:00pm EDT

Kalshi

● Fed maintains rate **95%** ● Cut 25bps **5%** ● Hike 25bps **2%**



Where we are and where we're headed

- Chapters 1-8 are about the long run
 - (Supply) Labor, capital, total factor productivity create **potential output** $\bar{Y}$
 - Social trends and human capital accumulation create a labor force
 - Given $\bar{Y}$, money and credit create a long-run rate of inflation, $\bar{\pi}$, because $MV = PY$
- Chapters 9-20 are (mostly) about the short run
 - (Demand) Expenditures can deviate from **potential output** $\bar{Y}$

$$Y = C + I + G + NX \lesseqgtr \bar{Y}$$

- A big reason is because **investment** responds to the real interest rate. Another is because all components respond to shocks. Those get amplified when consumption responds to current income
- When $Y > \bar{Y}$, the economy is booming, and $\tilde{Y} = \frac{Y - \bar{Y}}{\bar{Y}} > 0$ and the **Phillips curve** shows us $\Delta\pi > 0$. When the economy is slumping: the reverse

Summary of Class 15 on Chapter 10

- $MV = PY$ is useful for the long run and historical settings
- Modern central banking employs different tactics
- But especially in historical periods, $MV = PY$ reveals some important truths
- The money supply M is more than cash; it's really the amount of **credit** readily available
 - [Wikipedia on monetary aggregates](#); and a [11/12/24 Fed update](#)
 - $M0$ is what might first think of: coins and currency
 - $M1$ includes checking and saving deposits; $M2$ is less liquid still, etc.
 - **Banks are important for M**

Learning objectives for Chapter 11

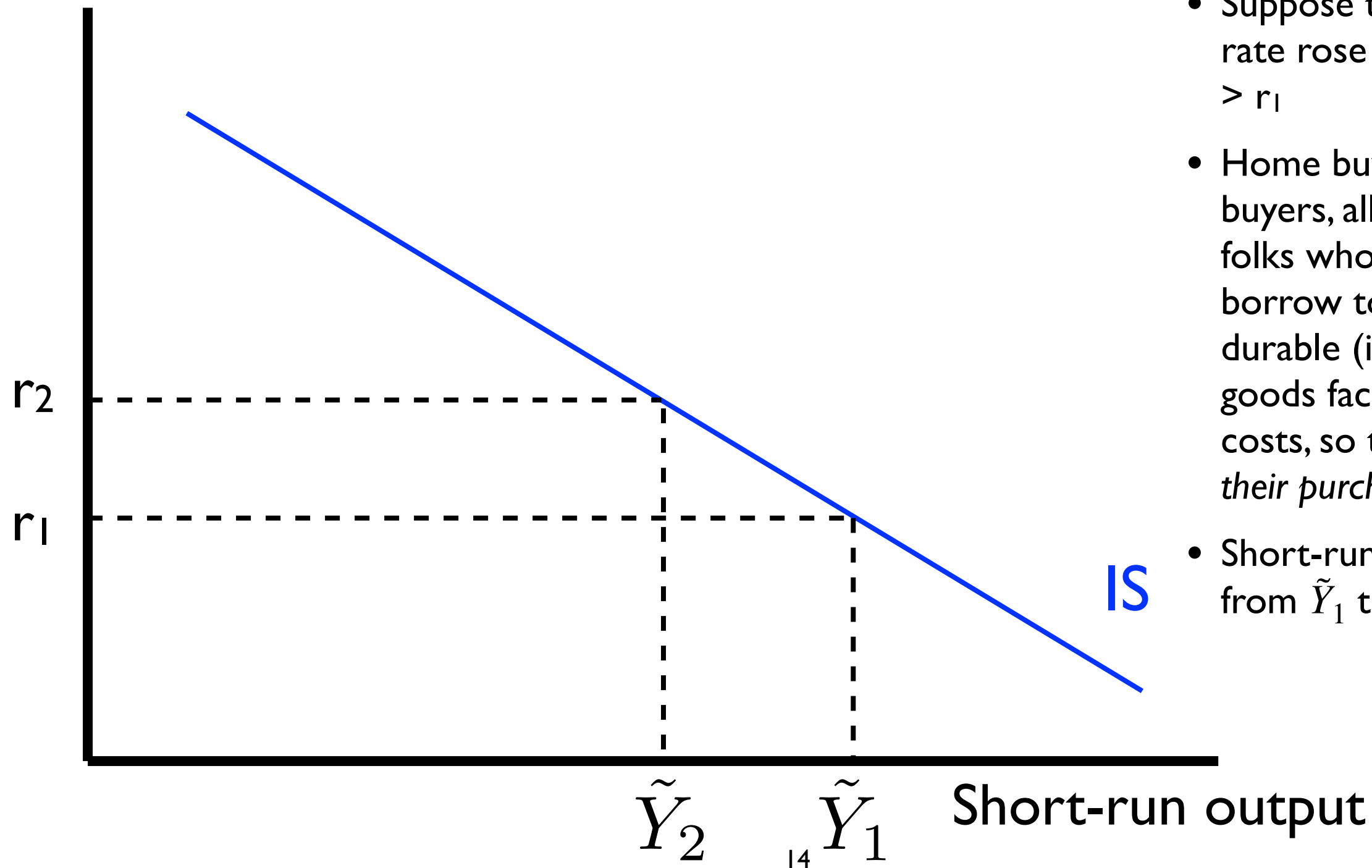
- Think of $Y = C + I + G + EX - IM$ as **aggregate demand**
- The IS curve shows how short-run output $\tilde{Y}$...
 - Is a downward-sloping function of the real interest rate R
 - Shifts inward when components of aggregate demand fall, and shifts outward when components of aggregate demand rise
- Several expenditure components might be sensitive to the interest rate, but we focus on **investment**
 - A higher real interest rate R is an increased price of spending today
 - (ECON 100A:) For many economic agents, the substitution effect of a higher R outweighs any income effect, and spending today declines when R rises
- The math of the IS curve is a downward sloping line: $\tilde{Y} = \bar{a} - \bar{b} (R - \bar{r})$

Making the IS Curve intuitive

- The IS Curve just formalizes this logic about the interest rate and the price of borrowed funds, and capital investment spending
- When the price of borrowed funds falls,
- The quantity of investment spending demanded increases
- And the same is true in reverse: If the interest rate were to rise, the quantity of investment would fall
- This is basically the same logic that we see describing a downward sloping demand curve
- And like a demand curve, the IS Curve also slopes downward!

The IS Curve shows activity decreasing when interest rates rise

Interest rate



- Suppose the interest rate rose from r_1 to $r_2 > r_1$
- Home buyers, car buyers, all kinds of folks who have to borrow to buy durable (investment!) goods face higher costs, so they *reduce their purchases*
- Short-run output falls from $\tilde{Y}_1$ to $\tilde{Y}_2$

How do we come up with this?

- We could just specify it, kind of like how a demand curve just always slopes down
- But there's a reason why a demand curve always slopes down: diminishing marginal utility or productivity
- Similarly, there's a reason why the IS Curve slopes down
- Let's begin by reviewing our notation:

Current output we observe

Y_t

Long-run trend or *potential* output

$\bar{Y}_t$

Short-run deviation from long-run potential output

$$\tilde{Y}_t = \frac{Y_t - \bar{Y}_t}{\bar{Y}_t}.$$

We're going to consider each component of GDP separately

- Why? Each component: Consumption, Investment, Government, Exports, Imports, *behaves a little differently over business cycles*, so we want to treat them separately

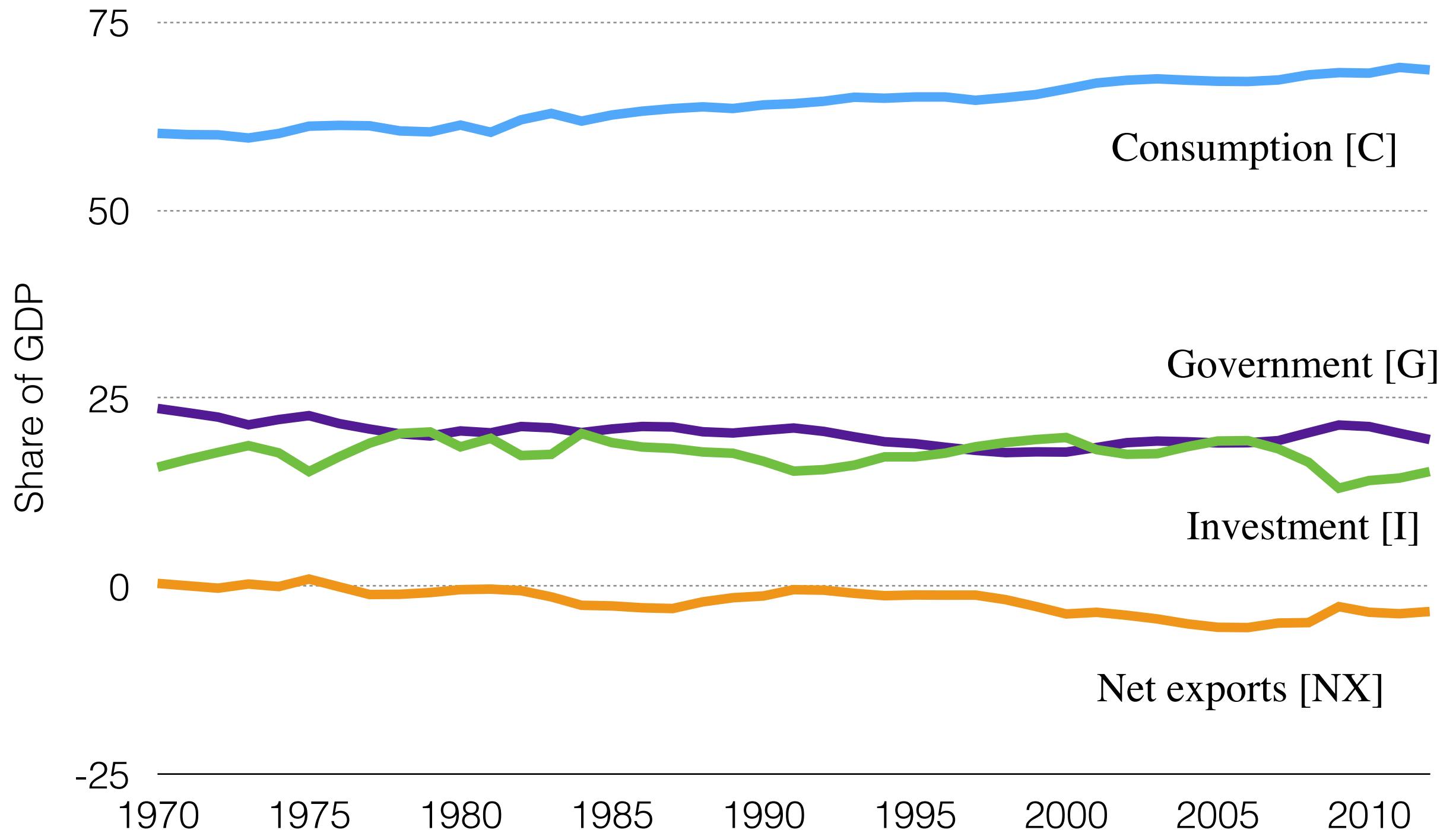
$$Y_t = C_t + I_t + G_t + NX_t$$

- Note that this is Y_t **actual** GDP, *including any fluctuations*
- We're now going to write out **EX**ports and **IM**ports also:

$$Y_t = C_t + I_t + G_t + EX_t - IM_t$$

- How do each of these behave? We're going to make most of them functions of potential (trend) income, $\bar{Y}_t$

Over time, the expenditure shares of GDP have been stable, with some drift, *except for investment*



Over time, the expenditure shares of GDP have been stable, with some drift, *except for investment*

- As a result, we will model each of the expenditure components *except for investment* as constant shares

- **Consumption** has averaged 64% of GDP since 1970

— Call that $a_c = 0.64$

- **Government** has averaged 20%

— Call that $a_g = 0.20$

- **Exports** have averaged 9%

— Call that $a_{ex} = 0.09$

- **Imports** have averaged 11%

— Call that $a_{im} = 0.11$

Investment has averaged 18% of GDP since 1970, *but it has jumped around a lot too*

so let's set $a_i = 0.18$

and let's model a part of investment that fluctuates around zero

Note that the a 's sum to 1

$$Y_t = \boxed{C_t} + I_t + \boxed{G_t} + \boxed{EX_t} - \boxed{IM_t}$$

$C_t = \bar{a}_c \bar{Y}_t$

$G_t = \bar{a}_g \bar{Y}_t$

$IM_t = \bar{a}_{im} \bar{Y}_t$

Each of these components
is just some fixed share of
potential GDP, $\bar{Y}_t$

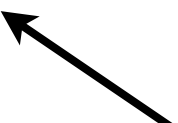
$$\boxed{EX_t = \bar{a}_{ex} \bar{Y}_t}$$

Why do we assume this?

Empirical grounds: Consumption is actually *smoother* than actual output, more like how potential output is smooth over time

Theoretical grounds: If you have savings, then when your income *temporarily* declines, you'll spend some savings to keep consuming

$$Y_t = C_t + I_t + G_t + EX_t - IM_t$$



$$\frac{I_t}{\bar{Y}_t} = \bar{a}_i - \bar{b} (R_t - \bar{r})$$

Investment is the special part of all this — it looks a lot different

What does it say? Suppose the right-hand part wasn't there:

$$\frac{I_t}{\bar{Y}_t} = \bar{a}_i \longrightarrow I_t = \bar{a}_i \bar{Y}_t \quad \text{Same format as before!}$$

The other part? When the *real interest rate*, R_t , is above the *marginal product of capital*, $\bar{r}$, then $R_t - \bar{r} > 0$ and investment drops **lower**

$$\frac{I_t}{\bar{Y}_t} = \bar{a}_i - \bar{b} (R_t - \bar{r})$$

- A numerical example to make this clearer:
- Suppose $\bar{a}_i = 0.15$ Then when $R_t = \bar{r}$, $I_t = 0.15 \bar{Y}_t$
- Now let $\bar{b} = 0.5$, and suppose $R_t = 10\%$, $\bar{r} = 15\%$

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- Then
$$\frac{I_t}{\bar{Y}_t} = 0.15 - 0.5 (10\% - 15\%)$$

$$\frac{I_t}{\bar{Y}_t} = 0.15 + 0.5 \times 5\%$$

$$\frac{I_t}{\bar{Y}_t} = 0.15 + 0.025 = 0.175$$

- Sensitivity to the interest rate increases investment *in the short run*; in the long run, investment must revert to $\bar{a}_i$

Deriving the IS Curve

$$Y_t = C_t + I_t + G_t + EX_t - IM_t$$

Divide through by
potential GDP

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Substitute for all
these using our rules

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$$\frac{Y_t}{\bar{Y}_t} = \underbrace{\bar{a}_c}_{\downarrow} + \underbrace{\bar{a}_i - \bar{b}(R_t - \bar{r})}_{\swarrow} + \underbrace{\bar{a}_g}_{\searrow} + \underbrace{\bar{a}_{ex}}_{\swarrow} - \underbrace{\bar{a}_{im}}_{\searrow}$$

Now subtract one
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Now subtract one
from both sides

$$\frac{Y_t}{\bar{Y}_t} - 1 = \frac{Y_t - \bar{Y}_t}{\bar{Y}_t} = \tilde{Y}_t$$

We'll have an equation for
the short-run fluctuation in
output!

What is the other side of the equation?

$$\tilde{Y}_t = \bar{a}_c + \bar{a}_i - \bar{b} (R_t - \bar{r}) + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} - 1$$

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$$\tilde{Y}_t = \bar{a}_c + \bar{a}_i - \bar{b} (R_t - \bar{r}) + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} - 1$$

Combine all the $\bar{a}_j$'s and the 1 and call it $\bar{a}$

$$\bar{a}_c + \bar{a}_i + \bar{a}_g + \bar{a}_{ex} - \bar{a}_{im} - 1 = \bar{a}$$

Then rewrite using $\bar{a}$

$$\tilde{Y}_t = \bar{a} - \bar{b} (R_t - \bar{r})$$

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Then rewrite using $\bar{a}$

$$\tilde{Y}_t = \bar{a} - \bar{b} (R_t - \bar{r})$$

This is the IS Curve, and it's just a straight line!

When the interest rate (R_t) is above the marginal product of capital (r),

(i) $R_t - r$ is positive; (ii) $-\bar{b}$ flips it negative; and (iii) $\tilde{Y}$ falls below zero

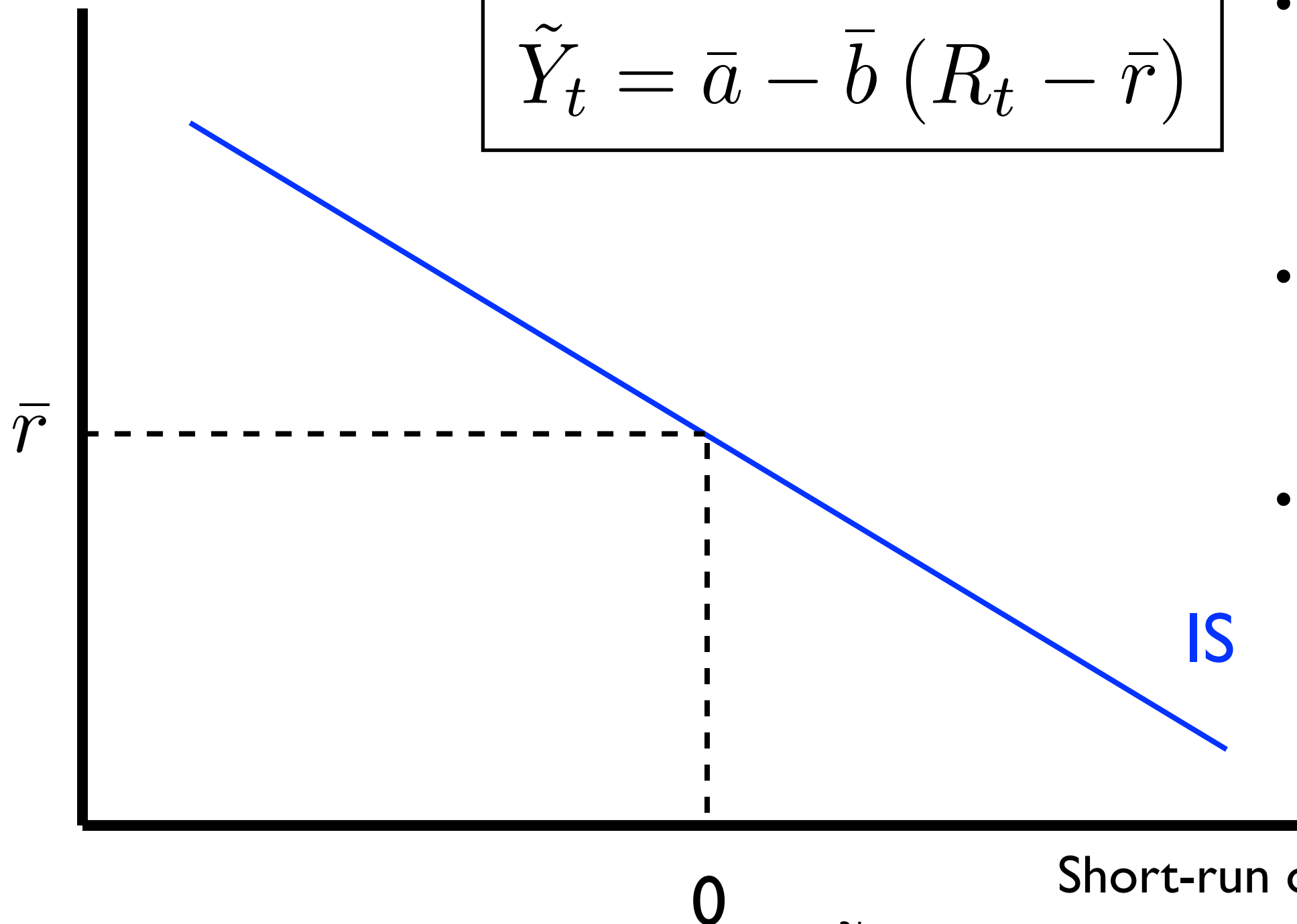
$$\tilde{Y}_t = \bar{a} - \bar{b} (R_t - \bar{r})$$

- What else does this say? In the **long run**, we know that $R = \bar{r}$ and $\tilde{Y} = 0$
- So in the long run, or when things are “normal,” that intercept $\bar{a} = 0$
- But there are times when $\bar{a}$ can change
- We call changes in $\bar{a}$ “demand shocks” because $\bar{a}$ derives from the demand side: consumption, investment, government spending, etc.
- Contrast this with shocks to prices or production, to technology, labor, or capital

What does this all look like? The same downward sloping curve we first proposed

Real interest rate, R_t

$$\tilde{Y}_t = \bar{a} - \bar{b} (R_t - \bar{r})$$

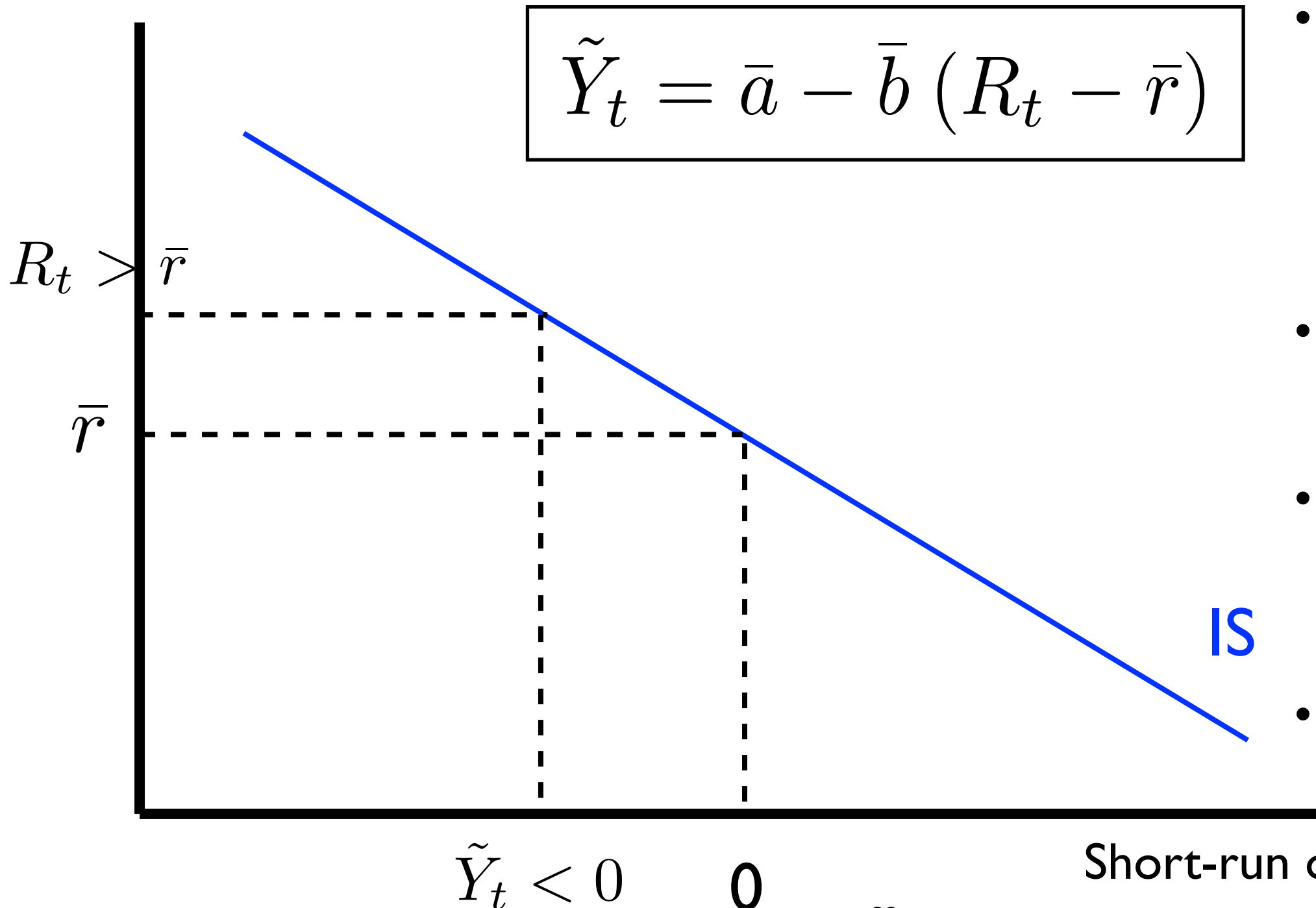


- The twist is that we're going to place the "zero" of the horizontal axis in the *middle* of the graph
- This where output is stable at its long-run potential and short-run output $\tilde{Y}_t = 0$
- Where does this occur? When the interest rate R_t is equal to the marginal product of capital, $R_t = \bar{r}$

What happens when the interest rate rises?

Real interest rate, R_t

$$\tilde{Y}_t = \bar{a} - \bar{b} (R_t - \bar{r})$$

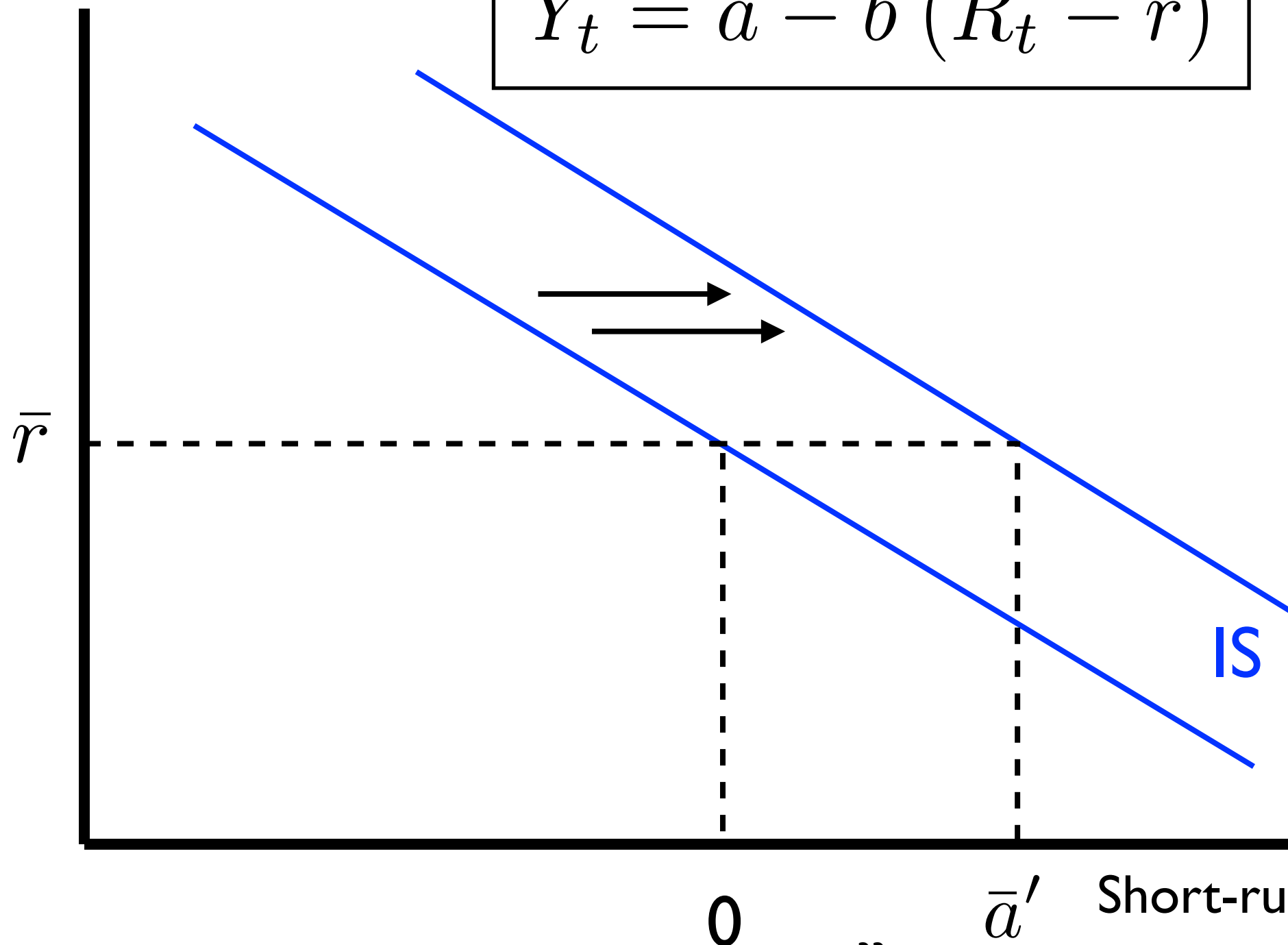


- If the interest rate rises above the marginal product of capital, **the IS Curve does not itself shift**
- Instead, this is a movement along the curve
- Investment activities like house purchases, capital expenditure *decline*
- Short run output becomes *negative*

What will **shift** the IS Curve itself? A change in the intercept term, $\bar{a}$

Real interest rate, R_t

$$\tilde{Y}_t = \bar{a} - \bar{b} (R_t - \bar{r})$$



- An **aggregate demand shock** will change the *intercept* of the IS Curve, $\bar{a}$
- Suppose people spend more than the usual share of potential income $\bar{Y}$ because the stock market is booming and they're feeling wealthier

$$C = \bar{a}_c \cdot \bar{Y}$$

- This is a story about $\bar{a}_c$ rising, which increases $\bar{a}$

i>clicker question 18.1

Suppose the interest rate R rises. What happens to $\tilde{Y}$?

- A. $\tilde{Y}$ will rise because of a rightward movement along the IS Curve
- B. $\tilde{Y}$ will fall because of a leftward movement along the IS Curve
- C. $\tilde{Y}$ will rise because of a rightward shift in the IS Curve
- D. $\tilde{Y}$ will fall because of a leftward shift in the IS Curve

i>clicker question 18.2

Suppose the interest rate R falls. What happens to $\tilde{Y}$?

- A. $\tilde{Y}$ will rise because of a rightward movement along the IS Curve
- B. $\tilde{Y}$ will fall because of a leftward movement along the IS Curve
- C. $\tilde{Y}$ will rise because of a rightward shift in the IS Curve
- D. $\tilde{Y}$ will fall because of a leftward shift in the IS Curve

i>clicker question 18.3

Suppose U.S. consumers raise their purchases because their houses appreciate in value. What happens to $\tilde{Y}$?

- A. $\tilde{Y}$ will rise because of a rightward movement along the IS Curve
- B. $\tilde{Y}$ will fall because of a leftward movement along the IS Curve
- C. $\tilde{Y}$ will rise because of a rightward shift in the IS Curve
- D. $\tilde{Y}$ will fall because of a leftward shift in the IS Curve

i>clicker question 18.4

Suppose U.S. government spending on goods & services falls because a fiscally conscious party takes control of Congress and the presidency and reduces spending.
What happens to $\tilde{Y}$?

- A. $\tilde{Y}$ will rise because of a rightward movement along the IS Curve
- B. $\tilde{Y}$ will fall because of a leftward movement along the IS Curve
- C. $\tilde{Y}$ will rise because of a rightward shift in the IS Curve
- D. $\tilde{Y}$ will fall because of a leftward shift in the IS Curve

i>clicker question 18.5

Why does the IS Curve slope downward?

- A. It slopes downward because as the interest rate R falls, investment spending rises
- B. It slopes downward because as the interest rate R falls, investment spending falls
- C. It slopes downward because the IS Curve is like a supply curve
- D. Trick question; it doesn't slope downward, it slopes upward